MIL-M-7255B(ASG)

6 OCTOBER 1969

Superseding
MIL-M-7255A(ASG)
17 January 1953

MILITARY SPECIFICATION

MANOMETER, PRESSURE, DIFFERENTIAL AND ABSOLUTE, TYPE E-1

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 This specification covers one type of differential and absolute pressure manometer, designated type E-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-B-626	Brass, Leaded and Nonleaded: Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5015	Connectors, Electric, AN Type
MIL-C-5541	Chemical Films and Chemical Film Materials, for Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standards, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

FSC 6685

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be address to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, Ill.,60606.)

3. REQUIREMENTS

3.1 Preproduction. - This specification makes provisions for preproduction inspection (see 4.4).

3.2 Materials. -

3.2.1 Metals. - Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed inclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment, anodizing of aluminum-alloy parts, and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments.

3.2.2 Protective treatment. - When materials are used in the construction of the manometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.2.3 Selection of materials. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in 3.1.3.1.

3.2.3.1 Standard parts. - Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, and so forth, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids or request for proposal, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction. -

3.3.1 The manometer shall be designed to apply, regulate, and indicate an absolute pressure simulating pressure altitude and a differential pressure simulating indicated airspeed independently of one another. The differential pressure shall use the controlled absolute pressure as the reference (low) pressure.

3.3.2 The manometer shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.3 Component parts. - The manometer shall consist of a manometer controller unit mounted on a base with provisions for leveling. The power supply units and necessary volume reservoirs for the efficient operation and control of the manometer may be mounted in the base. The manometer controller unit shall consist of a barometer unit, differential manometer unit, suppression manometer unit with the necessary controls, columns, scales, scanning devices for reading to the accuracy required, and an amplifier unit to control each barometer or manometer unit with the necessary switches for operating bubble levels and thermometer compensating devices for applying temperature and gravity corrections to the scales. This compensator device shall, after existing values of temperature and gravity are set in, automatically correct the readings for temperature and gravity error at all points on the scale throughout the entire range of the instrument. Convenient controls shall be provided for varying the absolute and differential pressures. Magnetic dump valves shall be provided in each system to return the respective system to atmospheric pressure when the main switch is in the "off" position. All connections between power supply units, amplifier units, barometer, and manometer units shall be inside the case. The connections to the vacuum and pressure source and the electric power supply shall be external on the side of the case.

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3.3.4 Range. - The range of the absolute pressure control shall be from 31 to at least 0.0425 inch Hg absolute (-1,000 to 150,000 feet). The range of differential pressure control shall be from 0 to 80 inches Hg. When this device is connected to airtight systems with volumes not exceeding 200 cubic inches, the control shall maintain the preset pressures within ±0.002 inch Hg ±50 percent at any point in the absolute pressure range and from 0 to 40 inches Hg on the differential pressure range. From 40 to 80 inches Hg on the differential pressure range, the tolerance shall be increased to 0.004 inch Hg.

3.3.5 Indication. - The manometers (absolute and differential) shall indicate in terms of inches of mercury at 0° C and standard gravity. The indication on all scales shall have incorporated in it a temperature gravity correction as specified in 3.3.11.

3.3.5.1 The range of indication of differential pressure shall be from 0 to 80 inches Hg correct to within 0.005 inch Hg from 0 to 40 inches, and to within 0.010 inch from 40 to 80 inches.

3.3.5.2 The range of indication of the absolute pressure shall be from 0 to 310 inches Hg absolute, correct to within 0.004 inch Hg from 31.0 to 1.0 inch Hg and to within 1/2 percent of the indication from 1.0 to 0.0425 inch Hg.

3.3.5.3 The manometer tubes may be filled with mercury or other stable liquid or a combination of both in order to obtain the required accuracy.

3.3.5.3.1 A sufficient quantity of liquid for each manometer tube shall be provided. The liquids shall be of the best laboratory quality.

3.3.6 Controls. - The controls for varying the absolute pressure and the differential pressure shall be designed for easy operation. The controls shall be designed so that the operator may readily select and adjust the absolute and differential pressures to any desired value within the range of the device. The actual regulating valves may be located remotely and shall be provided with 1/4-inch flared tube fittings for sources of pressure, vacuum, and outlets of static (absolute) and pitot (differential) pressure. The device shall be designed to operate on an air supply of approximately 80 pounds and the vacuum which can be obtained from a "Cenco Megavoc" vacuum pump, or equal. Consideration will be given to a different source of vacuum if required. All electrical devices shall be designed to operate on 115V 60-cycle ac. Provisions for a standard 2-3/4-inch round dial aircraft airspeed indicator shall be made adjacent to the differential pressure control and for a standard 2-3/4-inch round dial aircraft altimeter adjacent to the absolute pressure control. These instruments will be permanently connected to the output of the tester for rough checking on the operation of the tester. A test safety switch operating solenoid valves in both the pitot and static lines shall be provided for isolating the test equipment from the instruments being tested. This test safety switch shall be isolated from the other switches. In the event of power failure, the barometer, differential, and suppression systems shall fail in a "safe" condition. The main power switch shall in turn control the three amplifier units simultaneously. All switches shall be arranged for greatest simplicity and accessibility and shall be suitably marked to indicate their purpose.

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3.3.6.1 The scanners or reading devices on the barometer, differential, and suppressor units,shall operate smoothly throughout the full range of the scale and emplby suitable counter balances if need be to accomplish this. The gear ratio of the vernier reading mechanism shall be approximately 12 to 1.

3.3.7 Description. - Figure 1 shows a schematic diagram of a test apparatus which with proper instrumentation might satisfy the requirements of this specification. On figure 1, the pressure limiting switch is to turn off the sensitive barometer when pressures greater than its range are being used. If the barometer will not be damaged by pressures up to 31 inches Hg absolute, the valve may be omitted. Valves No. 9 and No. 10 are to isolate the sensitive manometer when its range is exceeded. The suppression pressure regulator is designed to maintain a pressure of 40 inches Hg differential between the regulated static pressure and the bottom of the suppression pressure manometer. When the change to a suppressed range is required, Valve No. 9 is closed thereby applying the 40 inches Hg differential to the top of the airspeed manometer. Valve No. 7 and No. 8 are to isolate the instruments being tested from the manometer. The foregoing does not contain any detail of the method of control and modification which may be required to make the system operate as desired. A drain line shall be provided from the overflow well to the lower cistern with a suitable valve at the overflow well.

3.3.8 Manometer tubes. - The tubes used to indicate pressures shall be constructed of glass having a low coefficient of expansion. The tubes shall be of good optical quality and shall have no flaws which in any way distort or obscure the view of the liquid column. The tubes shall have inside diameters of not less than 12 mm. The tubes shall have a uniform bore of the highest precision over the reading range.

3.3.8.1 Pressure limiting switches may be incorporated to prevent excessive pressure build-up in differential pressure system if necessary.

3.3.8.2 Suppression change-over shall be accomplished automatically and controlled by mercury level on the 40-80 column, to assure that the static system is not subjected to sudden pressure changes.

3.3.8.3 Floating type check valves shall be incorporated in each tube to prevent unloading pressures.

3.3.8.4 Dust free dehydrating dryers shall be incorporated in each of the manometer systems.

3.3.8.5 Solenoid operated restricted vent valves shall be incorporated to return the system to atmospheric pressure in the event of power failure. Restriction shall be sufficient to prevent damage to the system. Manually operated restricted vent valves shall be provided to return the instruments to atmospheric pressure in the event of power failure. The solenoid valve to the instruments (test safety) shall be of the "on-off+reset" type.

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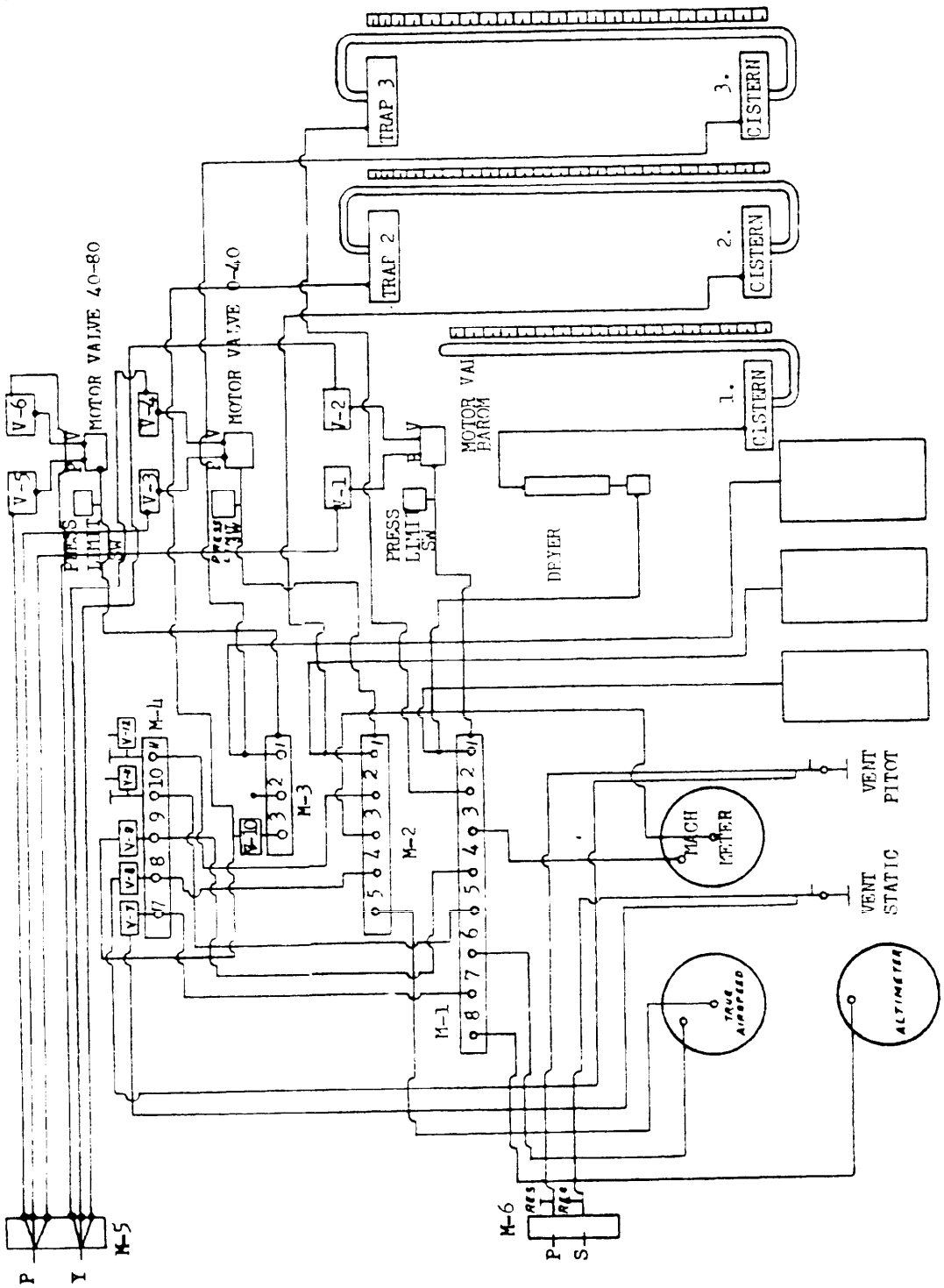


FIGURE 1. Test set-up

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3.3.8.6 Tube seal. - The top of the tubes used for measuring absolute pressure shall be closed by a manually operated, liquid sealed valve of a suitable type to permit renewal of the vacuum in the tube by evacuating the area above the seal through an incorporated vacuum connection and then applying a pressure to the well at the bottom of the tube. Suitable markings shall be provided to indicate the correct amount of liquid in the sealed valve. The valve and any lubricant used thereon shall be of such material as to prevent the contamination of the liquid. The valve shall be easily removable to permit cleaning the tube.

3.3.8.7 Liquid level. - Suitable marking shall be provided to indicate the correct amount of liquid in each tube.

3.3.9 Scales. - The scales shall, if they are not an integral part of the frame, be made of brass conforming to QQ-B-626, halfhard. The scales shall be attached to the frame in such a manner that any difference in expansion between the frame and scale will be automatically corrected. The scales shall have a linear and accurately known coefficient over the temperature range of 0° to +40° C. The scales shall be resistant enough to corrosion and wear to prevent the marking from becoming indistinct during the service life of the manometer.

3.3.9.1 Graduations. - The scales shall be graduated in inches of mercury and provided with a vernier so that it may be read to within 0.001 inch Hg. The scales shall be suitably foreshortened (not to exceed 2.5 percent) to automatically correct for the variation in liquid level in the well or cistern. The scale shall be graduated to indicate pressure in terms of a column of mercury at 0° C and gravity of 980.665 centimeters per second per second. The differential manometer scale shall be calibrated from 0 to 40 inches Hg and the suppression manometer scale shall be calibrated from 40 to 80 inches Hg. The scales shall be labeled accordingly. Scales shall also be calibrated in knots with indications at 5-knot intervals and numerals at each 50-knot interval throughout the full range of the scales. A magnifier of at least 4X shall be attached to the meniscus reading assembly which will permit reading of both mercury column and scale vernier.

3.3.9.2 Scale graduations. - Graduations on all scales and verniers shall be fine distinct lines not less than 0.003 or more than 0.005 inch wide. The height of the numerals on all scales and verniers shall be not less than 3/64 or more than 3/32 inch. The altitude scales shall be marked at each 1,000-foot interval from -1,000 to +100,000 feet and then at the 150,000- and the 200,000-foot range. The scale shall be subdivided as follows: -1,000 to +10,000 feet in 100-foot subdivisions; 10,000 to 50,000 feet in 500-foot subdivisions; 50,000 to 100,000 feet in 1,000-foot subdivisions; 100,000 to 150,000 feet in 10,000-foot subdivisions. No subdivisions shall be made in the 150,000- to 200,000-foot range.

3.3.10 Stop bars. - The barometer unit, the differential manometer unit, and the suppressor manometer unit shall each be provided with a stop bar. The barometer unit shall be provided with stops at 0, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 80 and 100,000-foot test points. The differential manometer unit shall be provided with stops at each 50-knot increment from 50 to 750 knots plus a stop at 40 inches Hg. The suppression manometer unit shall be provided with stops at 40, 60, 70, and 80 inches Hg, also at 800 and 850 knots.

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3.3.11 Temperature gravity corrector. - The temperature corrector shall be readily adjustable from 15° to 40° C. The gravity corrector shall be readily adjustable from 977 to 982 centimeters per second per second. Suitable scales shall be provided to indicate the value of temperature and gravity for which compensation is being applied. The gravity scale shall be divided into increments of two-tenths of one centimeter per second per second, with an additional graduation at standard gravity 980.665. Numerals shall be engraved on the temperature scale at 0, 10, 20, 30, and 40-degree points. Numerals shall be engraved on the gravity scale at the 977, 978, 979, 980, and 982 points. A scale shall be provided on the meniscus reading assembly to indicate the amount of correction being applied at any reading. The accuracy with which the temperature gravity corrector sets in the corrections at any point on the scales shall be determined by taking readings on the applied correction scale for various settings of the temperature and gravity indices. The error without regard to sign shall not exceed 0.004 inch Hg.

3.3.12 The amplifier and control units shall be partitioned and sufficiently insulated from the tube chamber to eliminate temperature changes after the temperature corrector has been set. Temperature variations in the manometer section shall not vary more than ±1° C.

3.3.13 All connecting tubing within the chambers shall be of stainless steel.

3.3.14 Fixed rates throughout the range of the system shall be incorporated in the manometer and manometer systems insofar as possible.

3.3.15 Proportioning rate valves shall be incorporated in the barometer and manometer systems. The proportioning rate valves shall be located behind the front panel and shall be readily accessible for adjustment, if necessary.

3.3.16 System vent valves shall be incorporated in the 0-to-40- and 40-to-80-inch cisterns as a precaution in the event of power failure.

3.4 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.5 Thermometer. - A thermometer shall be mounted in a metal frame and attached to the manometer frame. The thermometer bulb shall be located approximately 21 inches above the mercury level in the cistern. The thermometer shall have a range of at least 0° to 40° C with graduation marks every 1/2 degree. The thermometer shall be accurate to within 0.3° C over its range.

3.6 Over-all dimensions. - The manometer shall be rectangular in form approximately 88 inches high, 34 inches wide, and 17 inches deep. Toe room shall be provided in the base. This will require the use of a suppression device, or equivalent, to obtain differential pressure ranges up to 80 inches Hg.

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3.7 Inreads. -

3.7.1 Screw threads. - Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.7.2 Pipe threads. - Pipe threads shall conform to MIL-P-7105.

3.8 Electrical connectors. - The electrical connector used in the construction of the manometer shall be in accordance with MIL-C-5015, except in hermetically sealed instruments where a receptacle designed for hermetic sealing shall be used.

3.9 Performance. - The manometer shall be capable of meeting the requirements specified herein.

3.10 Finishes and protective coatings. -

3.10.1 Aluminum-alloy parts. - Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows (see 3.10.1.1 through 3.10.1.4).

3.10.1.1 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.10.1.2 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.10.1.3 Casting containing non-aluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.10.1.4 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.10.2 Steel parts. - Steel parts shall be cadmium-plated, where practicable, in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.11 Identification of product. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Holder for information. - A holder, or holders, shall be permanently mounted on the inside of the rear door of the case to contain the items indicated in 3.12.1 through 3.12.3, so that they will be easily accessible for operating personnel.

3.12.1 Calibration card. - A calibration card for the barometer and manometers shall be furnished. The residual errors obtained in the calibration test (see 4.5.1.3) shall be extended on the card opposite the readings selected as test points. The card shall be securely tied to the manometer before shipment.

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3.12.2 Assembly placard. - If necessary, one printed copy of instructions with diagrams and illustrations covering the set up and preparation for operation of the manometer shall be furnished. These instructions shall be mounted on a sheet of 1/8-inch clear plastic and shall not exceed 8 by 10 inches.

3.12.3 Correction chart. - A gravity correction chart shall be furnished by the contractor. The chart shall be approximately 4 by 5 inches in size. The chart shall be prepared in accordance with information specified on figure 2.

3.13 Workmanship. -

3.13.1 General. - The manometer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.13.2 Dimensions. - Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the manometer, they shall be held or limited accordingly.

3.13.3 Screw assemblies. - Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

3.13.4 Cleaning. - The manometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspections shall be classified as follows:

(a) Preproduction inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

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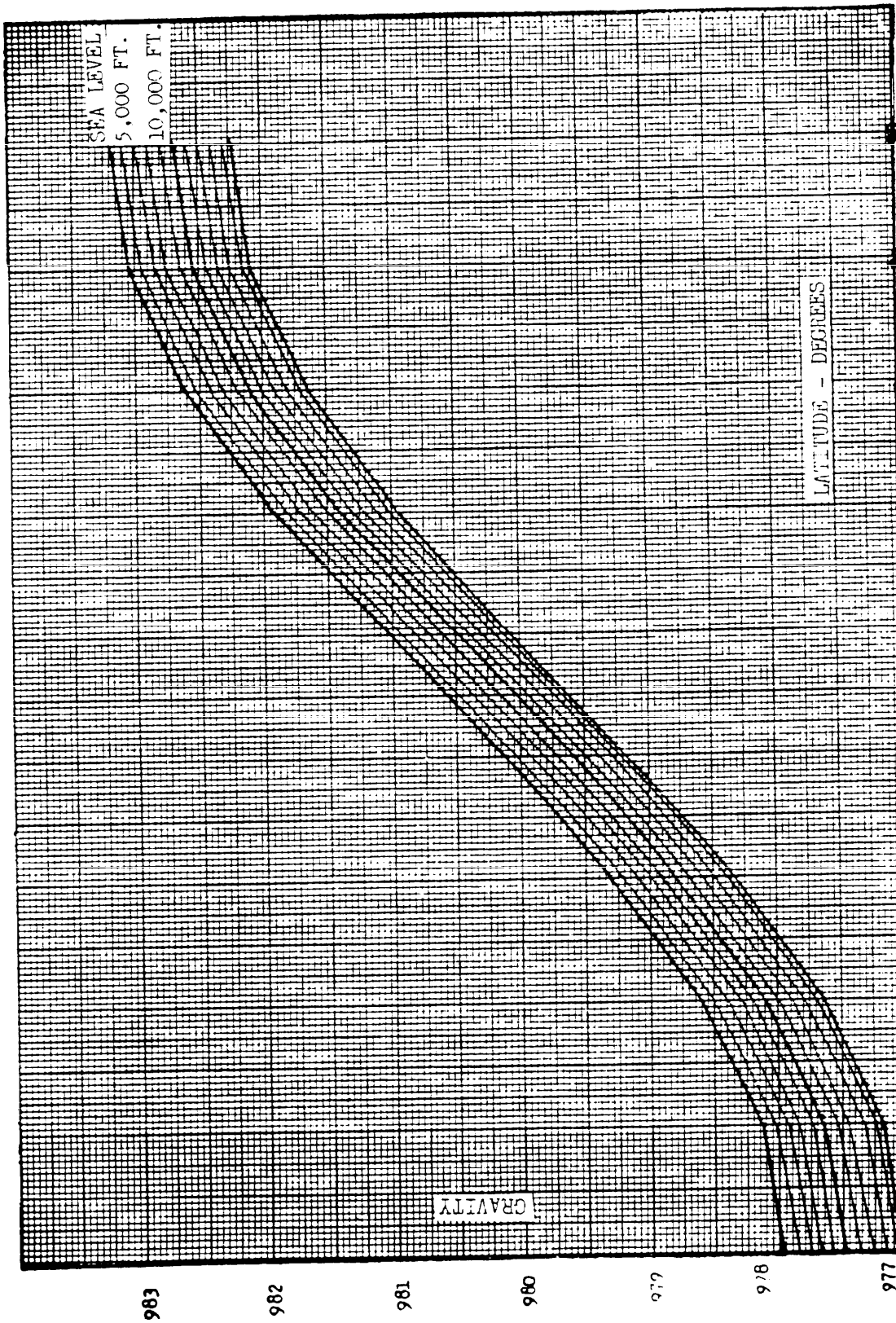


FIGURE 2. Chart - Barometer gravity correction

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4.3 Test conditions. -

4.3.1 Standard atmospheric conditions. - Whenever the pressure and temperature existing at the time of tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.2 Pressure measurement standard. - The contractor shall maintain two each secondary standards of a type and range suitable for calibrating the manometers specified herein. The standards proposed to meet this requirement shall be subject to the approval of the procuring activity.

4.3.2.1 Accuracy of pressure standards. - The standards selected shall be certified by the National Bureau of Standards within a 3-month period immediately preceding the testing of manometers specified in the contract. The secondary standards shall be certified by the National Bureau of Standards at intervals of not greater than 6 months.

4.4 Preproduction inspection. - The preproduction inspection shall consist of the examination and tests of this specification.

4.4.1 Sampling instructions. - One manometer, as specified in the contract or order (see 6.2), shall be tested and examined for design approval by the procuring activity, or when so specified in the contract, at the contractor's plant, under the supervision of the procuring activity.

4.5 Quality conformance inspection. - The quality conformance inspection shall consist of the individual tests.

4.5.1 Individual tests. - Each manometer shall be subjected to the examination and tests specified in 4.5.1.1 through 4.5.1.3.1.

4.5.1.1 Examination of product. - The manometer shall be inspected to determine compliance with this specification with respect to workmanship, material, dimensions, and markings.

4.5.1.2 Functional. - The manometer shall be operated to determine that component parts are properly assembled and to insure that the manometer functionally meets the requirements of section 3.

4.5.1.3 Calibration. - The manometer shall be tested for scale errors of the barometer unit and the differential pressure unit. The readings of the manometer shall be compared with pressures determined from simultaneous readings of the standards referred to in 4.3.2. The residual errors of the barometer unit shall not exceed 0.005 inch Hg at any point on the scale. Residual errors of the differential pressure unit shall not exceed 0.005 inch Hg from 0 to 40 inches, and 0.010 inch Hg from 40 to 80 inches.

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4.5.1.3.1 Calibration test points. - The barometer unit shall be tested at a minimum of 12 points as follows: From 31 to 6 inches Hg absolute, at intervals of not more than 3 inches; from 6 to 0 inches Hg absolute, at intervals of not more than 1 inch. The differential pressure units shall be tested at a minimum of 20 points as follows: From 0 to 40 inches Hg, at intervals of not more than 4 inches; from 20 to 80 inches Hg at intervals of not more than 4 inches.

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with this specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. -

5.1.1.1 Disassembly. - Items, components, and parts subject to loss or damage, or that would otherwise increase cube of the item at shipment, shall be disassembled. Removed nuts, bolts, and washers shall be secured to one of the mating parts. Parts removed, and mating parts remaining on the basic unit, shall be identified by stencilling with either letters or numbers, using contrasting color enamel. Removed parts too small to accommodate stencilling, shall have match marking identification indicated on tags meeting the requirements of MIL-STD-129. Disassembly shall not exceed the point where reassembly can be accomplished by unskilled labor with common tools.

5.1.1.2 Level A. - Manometers shall be preserved and packaged in accordance with MIL-P-116, Method III. The unit container shall conform to PPP-B-601, overseas type, style A or B.

5.1.1.3 Level C. - Manometers shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. The supplier may use his own commercial proactice, providing it meets this requirement.

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. - Manometers preserved and packaged as specified in 5.1.1 need not be overboxed. Cushioning, anchoring, bracing, and blocking within containers shall be in accordance with MIL-STD-1186.

5.2.2 Level B. - Same as for level A.

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5.2.3 Level C. - The manometer shall be packed in a shipping container which will be acceptable to the carrier at lowest rates and insure safe transportation to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers, as applicable to the mode of transportation.

5.3 Marking. - In addition to any other markings required by the order or contract (see 6.2), the interior and exterior markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The type E-1 manometer covered by this specification is intended for use in testing and calibrating aircraft Mach number indicators and true airspeed indicators.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Where the preproduction sample is to be forwarded for testing (see 4.4.1).
(b) Levels of packaging and packing required (see 5.1 and 5.2)
(c) Any special marking required (see 5.3).

6.3 Definitions. -

6.3.1 Residual error. - Residual error is defined as the difference between the true pressure and the observed indication after corrections have been set in by means of the mechanical correction device for the local value of gravity and ambient temperature.

6.4 Marginal indicia. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force 82

Preparing activity:
Air Force - 82

Project No. 6685-0318

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-M-[illegible](ASG) MANOMETER, PRESSURE, DIFFERENTIAL AND ABSOLUTE, TYPE E-1

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?
   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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the formation of frost/dew precipitation is clearly distinguishable. Frost/dew

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Aeronautical Standards Group

8719 Colesville Road

Silver Spring, Maryland 20910

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